

Technical Documentation

Autonomous Racing Vehicle — Hardware & Software Stack

29th ROBORACER Autonomous Racing Competition · IFAC 2026, Busan · Phase 0 Technical Materials

1. System Overview

Our platform is a 1/10-scale autonomous racing vehicle built on the ROBORACER (F1TENTH) standard. All sensing, computation, planning, and control run fully **onboard**; no off-board computation is used and no data is transmitted to the vehicle during a run. A 360° 3D LiDAR and a depth camera provide perception, an NVIDIA Jetson Orin Nano performs all real-time computation, and a VESC-driven powertrain propels a Traxxas 1/10 chassis. The autonomy software is organized as a pipeline: **3D LiDAR map-based localization · local trajectory planning · trajectory tracking & control · actuation**.

2. Hardware Stack

Component	Specification
Chassis	Traxxas 68086 Slash 4x4 Platinum Ultimate
Computer	NVIDIA Jetson Orin Nano, 8 GB memory (6-core CPU, 1024-core GPU, 40 TOPS)
LiDAR	Livox Mid-360 — 360° 3D LiDAR FOV 360° (H) × 59° (V), range 40 m @ 10% / 70 m @ 80%, 200,000 pts/s @ 10 Hz, built-in 6-axis IMU @ 200 Hz
Camera	Intel RealSense Depth Camera D455 RGB 1280×800 @ 30 fps (90° × 65°); Depth 1280×720 @ 90 fps (87° × 58°)
Motor Controller	VESC MK VI (input 11.1 - 60 V)
Motor	Traxxas Velineon 3500 brushless motor (max power 300 W)
Power Board	Matek UBEC Duo (Out1: 4 A, 5 - 12 V; Out2: 4 A, 5 V)
Battery	3S LiPo, 5000 mAh
IMU	Bosch BMI160 — 3-axis accelerometer + 3-axis gyroscope (6 DoF), built into VESC
Odometry	Provided by VESC (motor / ESC feedback)

3. Software Stack

The autonomy stack runs entirely onboard the Jetson Orin Nano and is composed of three subsystems — localization, local planning, and control — described below.

3.1 Localization

Vehicle pose is estimated by **map-based 3D LiDAR localization** using **hdl_localization** with the **fast_gicp** registration backend, against a 3D point-cloud map of the track built in advance.

- **hdl_localization** — real-time 3D LiDAR localization that fuses motion prediction from the LiDAR's built-in IMU in an Unscented Kalman Filter (UKF) with scan matching against the prebuilt 3D map, publishing the vehicle pose (odometry) in the map frame at sensor rate.

- **fast_gicp** — a high-speed point-cloud registration library providing multi-threaded GICP and voxelized VGICP (with optional CUDA acceleration). It is the scan-matching engine that aligns each incoming LiDAR scan to the map, yielding accurate, low-latency pose correction and drift-free, globally consistent estimates.

3.2 Local Planning

Local motion planning uses the **Timed Elastic Band (TEB)** planner. TEB represents the local path as a time-parameterized elastic band and optimizes it online for minimum traversal time while respecting the vehicle's kinodynamic limits (velocity, acceleration, turning radius) and maintaining clearance from detected obstacles. It outputs a smooth, dynamically feasible local trajectory and the velocity / steering reference that the control layer tracks.

3.3 Control

Trajectory tracking is handled by a **single controller, selected from two alternatives** based on track testing — either DACER++ or MPC. Only one of the two is active during a run:

- **DACER++ (reinforcement learning)** — a diffusion-based actor-critic *distributional* RL algorithm (*Enhanced DACER with Multimodal Q-value Distribution for Risk-Sensitive Stochastic Vehicle Environments*, IEEE IV 2025). It models the full state-action return distribution with a quantile-value network and quantile regression, enabling **risk-sensitive** policies that adapt between conservative and aggressive driving behavior under the stochastic, high-speed conditions of racing.
- **MPC (Model Predictive Control)** — a model-based controller that, at each step, optimizes steering and throttle commands over a finite prediction horizon to track the planned trajectory while enforcing actuator and vehicle-dynamics constraints.

The team runs whichever of the two controllers delivers better lap time and stability on the competition track; the two are interchangeable within the same localization and planning pipeline.

All computation is performed onboard the Jetson Orin Nano; no data is transmitted to the vehicle during normal driving, in compliance with the ROBORACER competition rules.